

Project Report

Learn and Contribute to Society





Final Project Report

Classifying Risk and Delay for TTC.

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Preface

Write your content.

Abstract

Type here the abstract of the project.

Chapter 1

Introduction

See this how you can add a reference, Knuth (1984).

1.1 Overview

1.1.1 Existing System

1.2 Objectives of Project (Must be clearly, precisely defined and must be covered in the work)

Chapter 2

Pre-Processing and Exploratory Data Analysis

Once you have your dataset, you need to preprocess it to make it suitable for machine learning. This includes cleaning the data, handling missing values, encoding categorical variables, and scaling numerical features. This step is crucial as the accuracy of your model will depend on the quality of your data.

2.1 Dataset Collection

The dataset we chose to use was the city of Toronto “TTC Bus Delay Data” 2025 dataset, as our objective for this project was aimed to relate to a real-world issue. The goal was to predict delay risk of a particular ttc vehicle on a specified day and also to predict the amount delay in minutes based off the datasets predictors and our own custom features. Finding the Dataset was simple as city of Toronto contains many publically available data that is regularly updated.

Here is the city of Toronto Dataset (2025) Dataset Source

2.2 Data Pre-processing

Some issues to consider to be included: - Are there missing values in your data? Is there any pattern in whether or not a particular value is missing? How will the missing values be dealt with?

- Are there problems because of data size (rows or columns)? Are some algorithms more affected? Can you come up with any solutions?
- If you are doing any clustering, are there clusters in the data? How many? How well separated are they? Do different methods agree on the number of clusters?
- Is scaling of the variables an issue? For what methods?
- Are there outliers in the data? How did you deal with them?

- Does transforming the data simplify any analysis?

2.3 Exploratory Data Analysis and Visualisations

For Exploratory Data analysis and visualizations we utilized visualizations to examine the relationship between our models. Specifically for the Regression models we plotted the predicted vs the actual data and a plot of the residuals to capture where our models struggled to find a relationship.

For the Classifications models we utilized tree plots to plot what the decision tree looked like after splits and applying kfold - cross-validation. Scatterplots were also utilized to plot the actual delay-risk compared to our models predicted delay-risk.

- Data Visualizations: The graphs we created were to understand the patterns and trends in the data. The provided visualizations tools:
 - Scatterplots were used to examine the relationship between two variables, the actual and the predicted.
 - Basic plot functions was used to examine a single model predictions or to plot decision trees.
 - We also utilized basic plot functions to compare our models before tuning vs after tuning. This was used for classification trees to see the accuracy before and after applying k-fold cross validation.
- Statistical analysis: The LinearRegression was used to analyze if the data had a linear relationship. To evaluate this we used MSE, RMSE and R^2 to evaluate meaningfully the error of our prediction of Delay in minutes and how much of the data it actually captures. For the DecisionTree Regression we evaluated using the same metrics as LinearRegression to compare the models. However with tree we saw significant benefits from hyperparameter tuning by running grid-search to find optimal parameters we saw lower error and a higher r^2 relationship. RandomForestRegressor also saw benefits from hyperparameters-tuning resulting in a lower-error and better R^2 . From comparing these models we learned that distribution of the data was not linear but non-linear likely due to significant outliers and variances within the data.
- Feature selection: For feature selection we used: Line, Day, Station, Code, Bound, Vehicle, Time_Group_late, night, Time_Group_early morning, Time_Group_noon, Time_Group_afternoon, Time_Group_evening, Time_Group_night, Day_Of_Week, Hour, Month and Rush_Hour. For the features Time_Group, Day, Hour, Month, Day_of_Week and Rush_Hour we used existing features to extract these features. As we concluded these would be meaningful and predictive to classifying risk and predicting the delay. We found these variables to be predictive, so we removed existing features after extracting more predictive features from them after feature engineering.
- Dimensionality reduction: (I didn't utilize this, can write about it if you did)

2.4 Other Related Sections (Optional)

Chapter 3

Methodology

3.1 Introduction to Python for Machine Learning

3.2 Platform and Machine Configurations Used

Google Colab and for machine configurations we used VS Code, Windows 11 os, Python 3.12.10, scikit-learn libraries, Panda, Numpy.

3.3 Data Split

Data splitting is the process of dividing the data into training, validation, and test sets. This step can help you evaluate the performance of the model on new data.

3.4 Model Planning

The next step is to select the appropriate machine learning or deep learning models (2 - 3) that can be used to solve the problem. This involves evaluating different models and selecting the one that is best suited for the problem.

3.5 Model Training:

This involves using the training data to adjust the model parameters to improve its accuracy.

3.6 Model Evaluation

Once the model is trained, the next step is to evaluate its performance on a validation dataset.

3.7 Model Optimization

Based on the results of the model evaluation, the next step is to optimize the model to improve its performance. This involves making changes to the model's architecture, tuning the hyperparameters, and retraining the model.

3.8 Final Model Building

Once the model is optimized, the final step is to fit the model and report final test error for each model

Present the information with support Screen Shots/ Figures (Each Figure must be numbered and Description of Figure must be provided)

Chapter 4

Results

To compare different machine learning models in the results section, here are some steps to consider:

4.1 Description of the Models

Start by providing a brief description of each model, including its purpose, algorithm type, and any relevant parameters or features used.

4.2 Performance Metrics

Describe the performance metrics used to evaluate the models, such as RMSE, MSE, accuracy, precision, recall, F1 score, or area under the receiver operating characteristic curve (AUC-ROC), etc based on the model class.

4.3 Results Table

Create a results table that presents the performance metrics for each model in a clear and concise format. The table should include the name of the model, the performance metric, and the value of the metric. It can also be helpful to include the standard deviation or confidence intervals for each metric.

4.4 Interpretation of the Results

Provide an interpretation of the results, highlighting any significant differences between the models. Discuss which model performed the best and why, as well as any limitations or potential sources of error.

4.5 Visualization

Provide visualizations to support the interpretation of the results, such as bar charts or box plots. These can help to illustrate the differences between the models and make the results more accessible to readers.

4.6 Sensitivity Analysis

Conduct a sensitivity analysis to test the robustness of the models. This can involve varying the parameters or features used in the models and re-evaluating their performance.

Chapter 5

Conclusion

Conclude by summarizing the key findings of the analysis, discussing the implications of the results, and suggesting areas for future research.

References

Knuth, Donald E. 1984. “Literate Programming.” *Comput. J.* (USA) 27 (2): 97–111. <https://doi.org/10.1093/comjnl/27.2.97>.